

Risk Methods — Quick Reference Card

VaR · ES · Backtests · Sensitivities · P&L attribution · Estimation · Stress · Capital. Deep companion: risk_methods_cheatsheet.pdf

◆ Universal Symbols

α	confidence (0.99 = 99%); tail = $1-\alpha$
s	sensitivity vector ($\partial V/\partial f_i$)
Σ	factor cov matrix (units must match s)
F	history matrix (rows=days, cols=factors)
$z(\alpha)$	norm.ppf(α); 99%=2.326, 99.5%=2.576
$N(\cdot), \varphi(\cdot)$	std-normal CDF, PDF

Sign: loss is POSITIVE (VaR, ES). P&L negative when you lose.

◆ Master Recipe (4 steps)

- Identify factors — rate, spread, vol, FX, equity (bucket KRD per tenor)
- Generate shocks — history / parametric ($z\cdot\sigma$) / MC (Cholesky)
- Revalue — sensitivity (linear) OR full-reval (exact)
- Aggregate — quantile (VaR) / mean-of-tail (ES) / gradient (MVar)

METHOD PICKER

Method	Shocks	Reval
Historical VaR	F (replay)	FULL
Parametric	Σ implied	sens (lin)
Monte Carlo	chol(Σ)-z	FULL
Filtered Hist	$F\cdot\sigma_t/\sigma_{\text{then}}$	FULL
Cornish-Fisher	$\Sigma + S, K$	sens (lin)

◆ Three patterns

P1 — SENSITIVITY-BASED

risk $\approx s\cdot\text{shock}$ or $\sqrt{(s^T\Sigma s)}$
Fast $O(\#\text{factors})$; falls down on non-linearity.

P2 — FULL REVALUATION

risk = $T(V(\text{shocked}) - V(\text{base}) \text{ over many shocks })$
Exact; cost = $n_{\text{paths}} \times \text{pricer cost}$.

P3 — EMPIRICAL DISTRIBUTION

risk = $T(\text{ empirical P\&L vector })$
No distribution assumption; bounded by window.

Historical = $P2+P3$; Parametric = $P1$; MC = $P2+\text{simulated}$; FHS = $P2+P3+\text{GARCH}$.

◆ Reference numbers (drill 13 portfolio)

Metric	Value
P&L std $\sigma = \sqrt{(s^T\Sigma s)}$	2.5327
Historical VaR(99.5%)	4.6781
Parametric VaR(99.5% / 99%)	6.5238 / 5.8919
MC VaR(99.5% / 99%) seed=42	6.3411 / 5.8760
Gammas (rate, spread, vol, fx)	1.86e-4, 4.29e-5, -1.06, 412.92
5y bond hist VaR(99.5%) [drill 8]	0.7363

4.68 (hist) < 6.34 (MC) < 6.52 (param): hist's bounded tail is THINNER than Gaussian; MC's full-reval captures gamma that param's linear misses.

● I — VaR & Expected Shortfall

Historical VaR (§1)

$VaR(\alpha) = -\text{sorted}(P\&L)[\lfloor (1-\alpha)\cdot N \rfloor]$
 $P3$ — empirical. No distribution assumption.

$\text{np.percentile}(pnl, (1-\text{conf})*100) \cdot -1$

Limits: bounded by window; vol-blind; calm year $\rightarrow$ tiny VaR.

Parametric / delta-normal VaR (§2)

$\sigma^2 = s^T \Sigma s$
 $VaR(\alpha) = z(\alpha) \cdot \sigma$

$P1$ — analytical Normal. Fast.

$\text{norm.ppf}(\text{conf}) * \text{np.sqrt}(s @ \text{cov} @ s)$

Assumes: factors Normal + book LINEAR. Misses fat tails + gamma.

Monte Carlo VaR (§3)

$L = \text{chol}(\Sigma)$; shocks = $z\cdot L^T \sim N(0, \Sigma)$
 $P\&L_k = V(\text{base} + \text{shock } k) - V_0$ (FULL reval)
 $VaR(\alpha) = -\text{percentile}(P\&L, (1-\alpha)\cdot 100)$

$P2$ + simulated. Captures gamma + custom distributions.

$L = \text{np.linalg.cholesky}(\text{cov})$
 $z = \text{rng.standard_normal}(n, 4)$
shocks = $z @ L^T$

Always set seed for reproducible risk numbers.

Expected Shortfall (CVaR) (§4)

Empirical: $ES\ \alpha = -\text{mean}(\text{sorted}[:, \lfloor (1-\alpha)N \rfloor])$
Normal: $ES\ \alpha = \sigma \cdot \varphi(z(\alpha)) / (1-\alpha)$
FRTB: ES at $\alpha=0.975$

Coherent (subadditive). Mean of tail, not boundary.

$k = \text{int}(\text{np.ceil}((1-\text{conf})*\text{len}(pnl)))$
 $-np.sort(pnl)[:k].\text{mean}()$

ES/VaR (Normal): $\alpha=0.99 \rightarrow 1.145\times$; $\alpha=0.975 \rightarrow 1.193\times$.

Filtered Historical Sim (FHS) (§5)

shock_today = ($\sigma_{\text{today}} / \sigma_{\text{then}}$) $\cdot F_i$
then proceed as historical VaR

$P2+P3+\text{GARCH rescale}$. Solves "stale during crises".

Pitfall: GARCH mis-spec $\rightarrow$ wrong rescale. Plot $\hat{\sigma}_t$.

Cornish-Fisher VaR (§6)

$z_CF = z + (z^2-1)S/6 + (z^3-3z)K/24$
 $- (2z^3-5z)S^2/36$
 $VaR = -(\mu + z_CF\cdot\sigma)$

Quick fat-tail fix without MC. $S=\text{skew}$, $K=\text{excess kurt}$.

Falls apart for $|S|>2$ or $K>10$ — use EVT/t-copula.

Marginal / Component / Incremental VaR (§7)

$MVaR_i = z(\alpha)\cdot(\Sigma w)_i / \sqrt{(w^T\Sigma w)}$
 $CVaR_i = w_i \cdot MVaR_i / [\Sigma CVaR_i = VaR \checkmark \text{ Euler}]$
 $IVaR_i = VaR(w) - VaR(w_{\{-i\}})$

M = per-\$ trade sizing; C = limit attribution; I = "if I close X".

z here = POSITION - return cov ($N\times N$), NOT factor cov. $\Sigma_pos = B\cdot\Sigma_factor\cdot B^T$.

Risk Budgeting / ERC (§7b)

solve $w : w_i\cdot(\Sigma w)_i / (w^T\Sigma w) = b_i \ \forall i$
ERC special case: $b_i = 1/N$
Maillard iteration: $w_i \leftarrow \text{target}/(w_i\cdot(\Sigma w)_i)$; renormalise
Inverse of \$7. SAA when expected returns uncertain. Risk-parity funds (Bridgewater).

$w_new = w * (\text{target} / (w * (\text{cov} @ w)))$
 $w_new /= w_new.\text{sum}()$

Stressed VaR (sVaR) (§8)

Historical VaR on FIXED 12M stress window
window chosen for MAX sVaR on CURRENT book
Basel II.5: Capital = $\max(VaR_t, k\cdot\text{avg}_{\text{60}})$
 $+ \max(sVaR_t, k\cdot\text{avg}_{\text{60}}\cdot sVaR)$

Example: calm 250d $\rightarrow$ VaR 2.3; 2008 window $\rightarrow$ sVaR 9.2 $\rightarrow$ capital $\sim 5\times$ higher.

▲ II — Backtesting

Kupiec POF (§9)

$LR = 2[x\cdot\ln(p/\pi) + (T-x)\cdot\ln((1-p)/(1-\pi))]$
 $\sim \chi^2(1)$; reject if $LR > 3.841$
 $\pi = 1-\alpha$, $p = x/T$, $x = \#\text{hits}$

Unconditional hit-rate test. Ignores clustering.

Christoffersen independence (§10)

$LR_ind \sim \chi^2(1)$ on 2×2 transition table
 $LR_CC = LR_POF + LR_ind \sim \chi^2(2)$

Catches vol-blind model: 6 hits OK count but all in 1 week.

Basel traffic light (§11)

Hits	Zone	Multiplier
0-4	GREEN	3.00
5	YELLOW	3.40
6	YELLOW	3.50
7	YELLOW	3.65
8	YELLOW	3.75
9	YELLOW	3.85
≥ 10	RED	4.00

99% 1-day VaR over 250d. Capital = $\text{avg_60d} \cdot k$.

ES backtest (Acerbi-Szekely T2) (§12)

$Z = (1/T)\cdot\Sigma[X\cdot I / (ES\cdot(1-\alpha))] - 1$
 $H_0: E[Z]=0$; $Z < 0$ (negative) = under-est
Mandated by FRTB IMA. Low power at $\alpha=0.975$ (~ 6 hits / 250d).

■ III — Risk Sensitivities

First-order (DV01, CS01, vega, FX-Δ, IE01) (§13)

$s_f = [V(\text{base} + \delta_f) - V(\text{base})] / \delta_f$
 $\Delta V \approx \Sigma_f s_f \cdot \text{shock_f}$

Greek	Factor	Bump
DV01	parallel rate	+1 bp
CS01	credit spread	+1 bp
IE01	breakeven inflation	+1 bp
FX-Δ	FX shock	+1% (0.01)
Vega	implied vol	+1 vol pt (0.01)

Convention: "DV01 = $V_{\text{bumped}} - V_{\text{base}}$ " $\rightarrow$ long bond DV01 is NEGATIVE.
The other convention (loss-given-up) flips sign. Pick one and LABEL it.

Second-order (gamma, convexity, cross-gamma) (§14)

$g_f = [V(+\delta) - 2V_0 + V(-\delta)] / \delta^2$ own
 $g_{\{ab\}} = [V(+a+b) - V(+a-b) - V(-a+b) + V(-a-b)] / (4\cdot\delta_a\cdot\delta_b)$ cross

Drill 12 gammas: (1.86e-4, 4.29e-5, -1.06, 412.92). Adding 2nd-order shrank residual 4 $\times$.

Key Rate Duration (KRD) (§15)

$KRD_k = V(\text{bump } z(T_k) + 1\text{bp}) - V(\text{base})$
 $\Sigma KRD_k \approx \text{parallel DV01}$

Curve-shape risk. Hedge vector, not scalar.

Pension fail mode: 5y IRS hedging 30y liab = parallel-neutral, drowns in steepener.

Computation methods (§16)

Method	Cost
Forward FD	$1\cdot n_{\text{factors}}$
Central FD	$2\cdot n_{\text{factors}}$ ($O(\delta^2)$)
AAD (adjoint)	$\sim 3\cdot 5\times 1$ fwd pass for ALL

FRTB SBM with 1000s of buckets $\rightarrow$ AAD pays off above ~ 50 factors.

Black-76 Greeks (forward F) (§17)

Δ	$DF\cdot N(d1)$
Γ	$DF\cdot\varphi(d1) / (F\cdot\sigma\cdot\sqrt{T})$
v	$DF\cdot F\cdot\varphi(d1)\cdot\sqrt{T}$
θ	— for long opt (time decay)

Γ uses **F** not **S** in denominator. Free from price formula.

◆ IV — P&L Attribution

Greek-based Taylor explain (§18)

```
first_f = s_f · Δf
second_f = ½ · g_f · Δf²
unexp = full - Σfirst - Σsecond
```

Drill 12 result: full −3.024 / first −3.054 / 2nd +0.023 / unexp +0.008.

Rule of thumb: |unexp| < 5% |full| for vanilla derivs.

Risk-factor explain (RFE) (§19)

```
Sens RFE: contrib_f = s_f · Δf (order-indep, misses cross)
Full RFE: contrib_f = V(only Δf) - V(base) (+ residual)
Sequential: apply IN ORDER; Σ = full, order-dep
```

Vanilla → sens. Options → full + cross-gamma report. Regulator → sequential.

Actual / Hyp / RT P&L (§20)

```
Actual = V_eod(pos_eod, mkt_eod) - V_sod(pos_sod, mkt_sod)
Hyp = V(pos_sod, mkt_eod) - V(pos_sod, mkt_sod)
RT (FRTB) = V_VaR_pricer(pos_sod, ...)
Unexp_intraday = Actual - Hyp
PLA-residual = Hyp - RT
```

Two residuals: desk behaviour (unexp_intraday) vs model quality (PLA-residual).

FRTB PLA test (§21)

```
p_s ≥ 0.80 / 0.70 / <0.70 = G/A/R
KS < 0.09 / 0.12 / ≥0.12 = G/A/R
Joint zone = WORST of the two
```

RED → desk excluded from IMA → SA capital fallback.

```
| spearmanr(hyp, rt); ks_2samp(hyp, rt)
```

Carry / pull-to-par / theta (§22)

```
Time_PnL = V(mkt_today, t+1) - V(mkt_today, t)
          = carry + pull_to_par + theta + cross-roll
```

"What would have happened with NO market move." Carry is EXPECTED daily P&L.

● V — Estimation Inputs

Historical vol (§23)

```
σ² = (1/(N-1)) Σ(r - r̄)²
σ²annual = σ²daily · √252
```

Equal weights. Stale during regime shifts.

EWMA (RiskMetrics) (§24)

```
σ²_t = λ · σ²_{t-1} + (1-λ) · r²_{t-1}
λ = 0.94 daily, 0.97 monthly
half-life = ln(0.5)/ln(λ) = 11.2d @ 0.94
```

One parameter; vol updates immediately. No mean reversion.

GARCH(1,1) (§25)

```
σ²_t = ω + α · r²_{t-1} + β · σ²_{t-1}
σ²_∞ = ω/(1-α-β), α+β < 1 stationary
α+β = 1 ≡ EWMA (IGARCH)
```

Adds mean reversion. Forecast σ̂_{t+h} → σ_∞.

```
| arch_model(r*100, vol="GARCH", p=1, q=1).fit()
```

Implied vol (§26)

```
| Solve σ: BS(σ) = market_price
| Brent on f(σ)=BS(σ)-mkt; σ ∈ [1e-6, 5]
```

Forward-looking; surface = vol by strike x expiry. Use Brent over Newton at boundaries.

Sample covariance (§27)

```
| Σ^= cov(R, rowvar=False, ddof=1)
Gold standard ONLY when T >> N. T < N → singular.
```

Ledoit-Wolf shrinkage (§28)

```
| Σ^shrink = δ · F + (1-δ) · S
δ* = max(0, min(1, π/γ²))
Regularises Σ when N≈T. δ ≈ 0.1-0.4 for daily equity.
| LedoitWolf().fit(R).covariance_
```

PSD repair (§29)

```
| Σ = QΛQᵀ; clip λ_i to ε
Σ_psd = Q·diag(max(λ,ε))·Qᵀ
rescale diagonal to preserve variances
```

Standard fix when cov is "almost-PSD" (numerical noise, missing data, bumps).

▲ VI — Stress & Scenarios

Historical scenario library (§30)

Event	Headline shocks
Black Monday '87	S&P −22%, vol x3
LTCM '98	EMBI +800bp, USTs flight
Lehman week '08	Ligor-OIS +200, S&P −12, USTs −80
COVID Mar '20	S&P −34, VIX 80, IG +250, HY +600
Gilt LDI '22	30y +120bp/3d, GBP −5

Hypothetical templates (§31)

```
| Parallel ±100bp · 2s10s steepener/flattener
| IG +200bp / HY +500bp blowout
| vol +50% relative · FX ±10% (3σ)
| risk-off combo: rate −50, spread +100, vol +30%
```

Solvency II SCR uses prescribed 1-in-200y deterministic shocks.

Reverse stress test (§32)

```
| find s* minimising sᵀΣ⁻¹s subject to PnL(s) = −L*
| Linear closed-form: s* = (−L*/sᵀΣs) · Σ · sens
| Mahalanobis = L* / σ_pnL
```

"Find cheapest way to lose \$X". Hedged book = high Mahalanobis. ICAAP requirement.

Aggregating scenarios (§33)

```
| Report: worst case ; full table
| DON'T: average / sum
```

Why no sum: Lehman + COVID both contain equity shock → adding implies it happens TWICE. Overstates loss 50-200%.

◆ VII — Capital Frameworks

Basel SA-CCR (§34)

```
| EAD = 1.4 · (RC + PFE)
| RC = max(V-C, 0)
| PFE = mult · Σ Add0n_AC
| mult = min(1, 0.05 + 0.95 · exp((V-C)/(2 · Σ Add0n)))
```

Replaces CEM. Counterparty exposure for OTC + SFTs.

Basel IMA legacy (§35)

```
| MRC = max(VaR_t, k · avg_σσVaR)
| + max(sVaR_t, k · avg_σσsVaR)
| k ∈ [3.0, 4.0] (traffic light)
```

Phased out by FRTB Jan 2025. Still used in some emerging-market regulators.

FRTB SA (SBM) (§36)

```
| Delta: √( Σ WS_b² + Σ {b≠c} p_bc · WS_b · WS_c )
| Vega: same on vega sens
| Curvature: max(CVR_up, CVR_down)² aggregated
| + DRC + RRA0
```

Sensitivity-tree: 5 risk classes x dozens of buckets x per-bucket aggregation.

FRTB IMA (§37)

```
| IMCC = 0.5 · ES_full_stressed + 0.5 · ES_reduced
| ES at α=0.975 with LH scaling
| IMA = IMCC + SES (NMRF) + DRC
```

ES(97.5) ≈ VaR(99) under Normal. Desk-level PLA + backtest required.

Operational cost: most banks dropped IMA for many desks; SA fallback.

IRC / DRC (default risk) (§38)

```
| IRC: MC on Vasicek ASRF, 99.9%/1y
| DRC: JTD_i = LGD-notional - MtM
| DRC = Σ buckets max(Σlongs, |Σshorts|)
```

Default = jump, missed by VaR. Concentrated single-names get hammered.

◆ Coherence axioms (Artzner)

#	Axiom	VaR	ES
1	Monotonicity	✓	✓
2	Translation invariance	✓	✓
3	Positive homogeneity	✓	✓
4	Subadditivity	✗	✓

VaR can punish diversification. Two indep low-prob big-loss assets: indiv VaR(0.95)=0, joint VaR(0.95)>0. ES never. FRTB chose ES for this reason.

Use: ES for capital, VaR for limits.

◆ Risk-factor sign cheatsheet

Long position	DV01	CS01	IE01	FX-Δ	Vega
Vanilla bond	−	−	0	0	0
Linker	−	0	+	0	0
Foreign bond	−	−	0	+	0
FRN (between resets)	≈0	−	0	0	0
IRS receive-fixed	−	0	0	0	0
IRS pay-fixed	+	0	0	0	0
ZCIS receive-infl	≈0 par	0	+	0	0
FX Spot/Fwd (long for)	+ small	0	0	+	0
CDS protection BUYER	− small	+	0	0	0
CDS protection SELLER	+ small	−	0	0	0
Long call (FX/swaption)	small	0	0	+Δ	+
Long put	small	0	0	−Δ	+
Pension annuity liab (short)	+	0	−	0	0

Vendor convention: Bloomberg / risk-systems DV01 is LOSS-given-up (long bond = +). FLIP every sign in DV01 column above before plugging into s · Δr.

◆ Top 6 traps (6-month memory)

- Units must match between s and Σ (bp vs decimal vs %). Always check σ_pnL in book units.
- DV01 sign convention — pick ∂V/∂r or loss-given-up; label EVERY report.
- 250-day window at α=0.995 → VaR = 2nd worst day → enormous sampling error.
- Gamma uses F (forward), not S (spot) in Black-76 denominator.
- Always set MC seed — risk numbers must be reproducible day-to-day.
- VaR not subadditive → can punish diversification on heavy-tail / discrete-loss books. Use ES for capital.